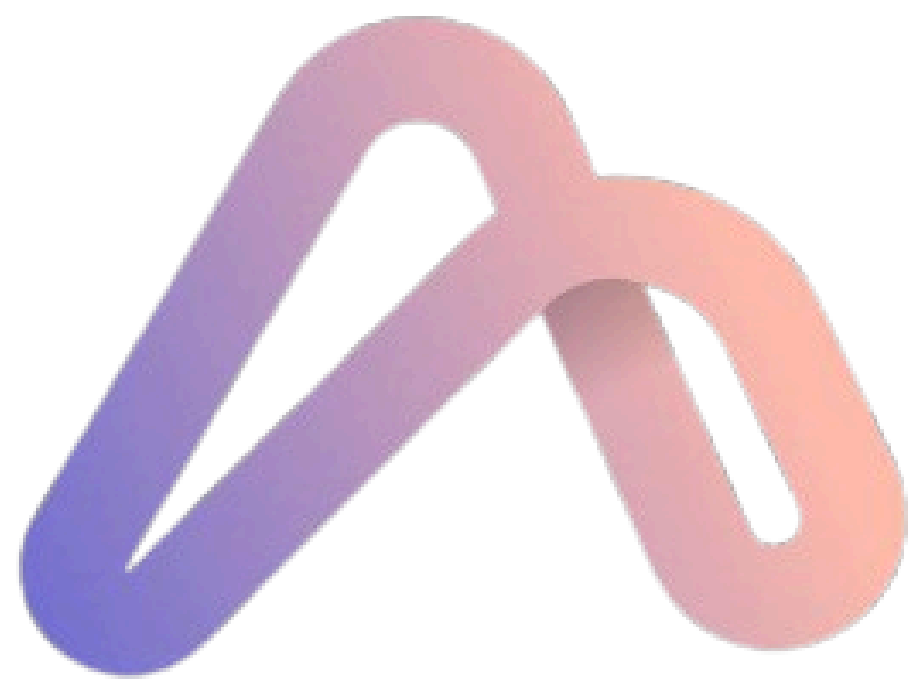




Levenshtein Distance

For Azerbaijan Artificial Intelligence Lab

Sabuhi Nazarov

We will cover

- Levenstein Distance
 - Applications
 - Operations
 - Formula
- Recursive approach
- Dynamic Programming approach
 - DP Levenstein distance Table
 - BackTrace

Are the two strings similar?
How similar?

algorithm

logarithm

a	l	g	o	r	i	t	h	m
↓	↓	↓	↓	↓	↓	↓	↓	↓
×	×		×					
↓	↓	↓	↓	↓	↓	↓	↓	↓
l	o	g	a	r	i	t	h	m

How we can measure this
similarity(or difference)?

Levenshtein Distance

Minimum number of editing operations
required to transform one string into the
other.

Can levenstein distance be
used for...?

Spell Correction

Oh, I accidentally typed **“reieve”**

Which word is the closest?

receive

recital

relieve

recede

Comparing DNA Sequences

GTACGTACGTA

VS

GATCGTCGTA

Are they relative?
Let's align them.

GTACGTACGTA

G_ATCGTCGTA

Let's see
Levenshtein Distance
operations...

ab --> abc

Insertion (adds a
character)

abc --> ab

Deletion (removing a
character)

ab --> ac

Substitution (replacing
one character with
another)

Note: Each operation has a cost of 1 (without weighting).

Real example?

algorithm → **logarithm**

1. Replace (o -> a)

algorithm → al**a**rithm

2. Delete (a)

algorithm → lgorithm

3. Add (o)

lgorithm → lo**a**rithm

algorithm $\longleftrightarrow$ **logarithm**

We found that levenstein distance
between these two strings is 3

Do you enjoy math?

Levenstein Distance formula

$$\text{lev}_{a,b}(i, j) = \begin{cases} \max(i, j) & \text{if } \min(i, j) = 0, \\ \min \begin{cases} \text{lev}_{a,b}(i-1, j) + 1 \\ \text{lev}_{a,b}(i, j-1) + 1 \\ \text{lev}_{a,b}(i-1, j-1) + 1_{(a_i \neq b_j)} \end{cases} & \text{otherwise.} \end{cases}$$

Looks complicated?
what about now...

$$\text{lev}_{a,b}(i, j) = \begin{cases} \text{max}(i, j) & \text{if } \min(i, j) = 0, \\ \min \begin{cases} \text{lev}_{a,b}(i-1, j) + 1 \\ \text{lev}_{a,b}(i, j-1) + 1 \\ \text{lev}_{a,b}(i-1, j-1) + 1_{(a_i \neq b_j)} \end{cases} & \text{otherwise.} \end{cases}$$

START HERE

do this

DO THIS

Recursive Approach vs Dynamic Programming Approach

Time Complexity

Naive Recursive

$$O(3^{\min(m,n)})$$

Dynamic Programming

$$O(m * n)$$

Naive Recursive follows the brute-force approach by considering all three possible operations at every step. Dynamic programming optimizes this process by storing intermediate results in a table(matrix).

Dynamic Programming Levenshtein Distance Table

weather **————→** **whether**

Levenstein distance?

	#	w	e	a	t	h	e	r
#	0	1	2	3	4	5	6	7
w	1							
h	2							
e	3							
t	4							
h	5							
e	6							
r	7							

	#	w	e	a	t	h	e	r
#	0	1	2	3	4	5	6	7
w	1							
h	2							
e	3							
t	4							
h	5							
e	6							
r	7							

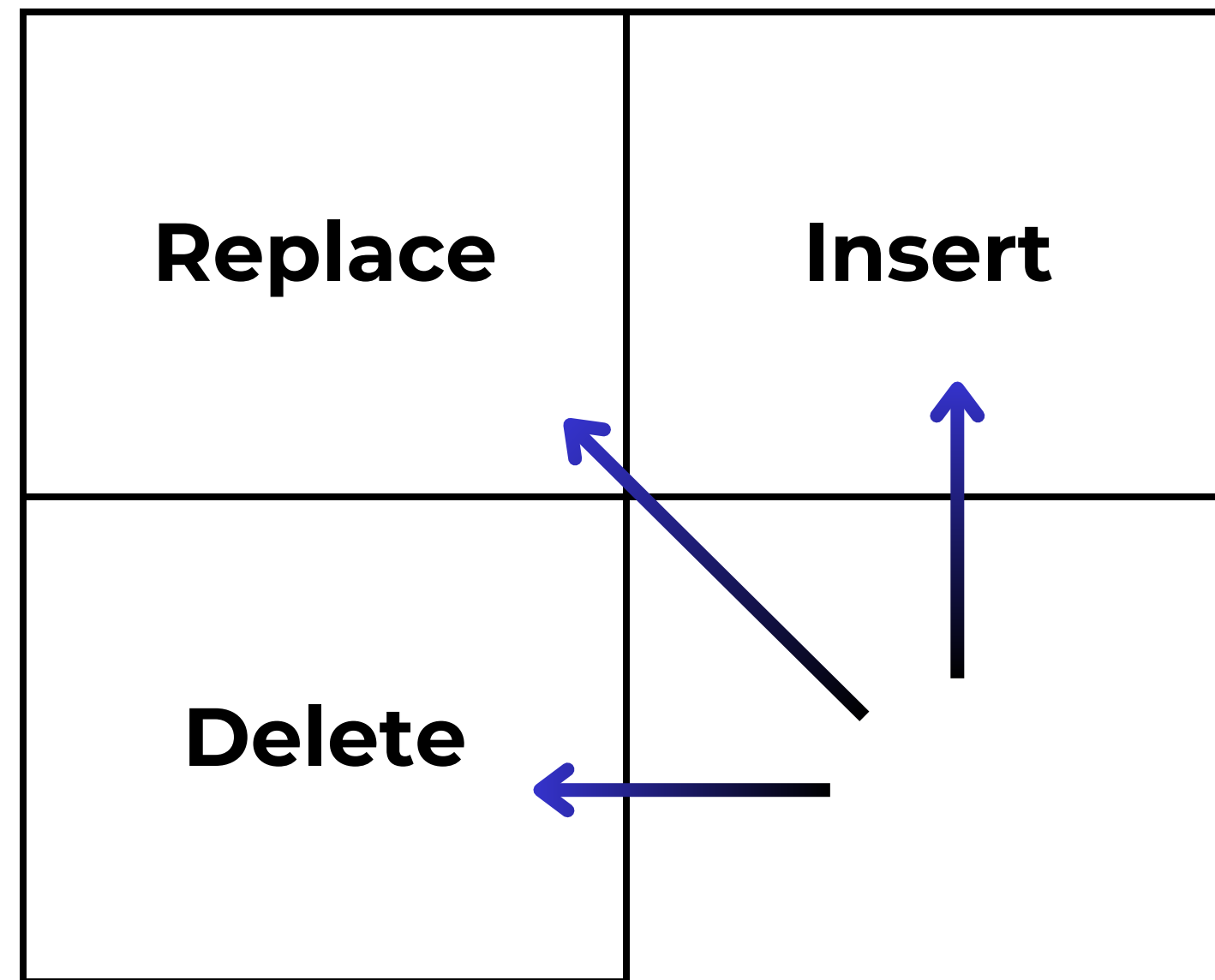
$$\text{lev}_{a,b}(i, j) = \begin{cases} \max(i, j) & \text{if } \min(i, j) = 0, \\ \min \begin{cases} \text{lev}_{a,b}(i-1, j) + 1 \\ \text{lev}_{a,b}(i, j-1) + 1 \\ \text{lev}_{a,b}(i-1, j-1) + 1_{(a_i \neq b_j)} \end{cases} & \text{otherwise.} \end{cases}$$

	#	w	e	a	t	h	e	r
#	0	1	2	3	4	5	6	7
w	1	0	1	2	3	4	5	6
h	2	1	1	2	3	3	4	5
e	3	2	1	2	3	4	3	4
t	4	3	2	2	2	3	4	4
h	5	4	3	3	3	2	3	4
e	6	5	4	3	3	3	2	3
r	7	6	5	4	4	4	3	2

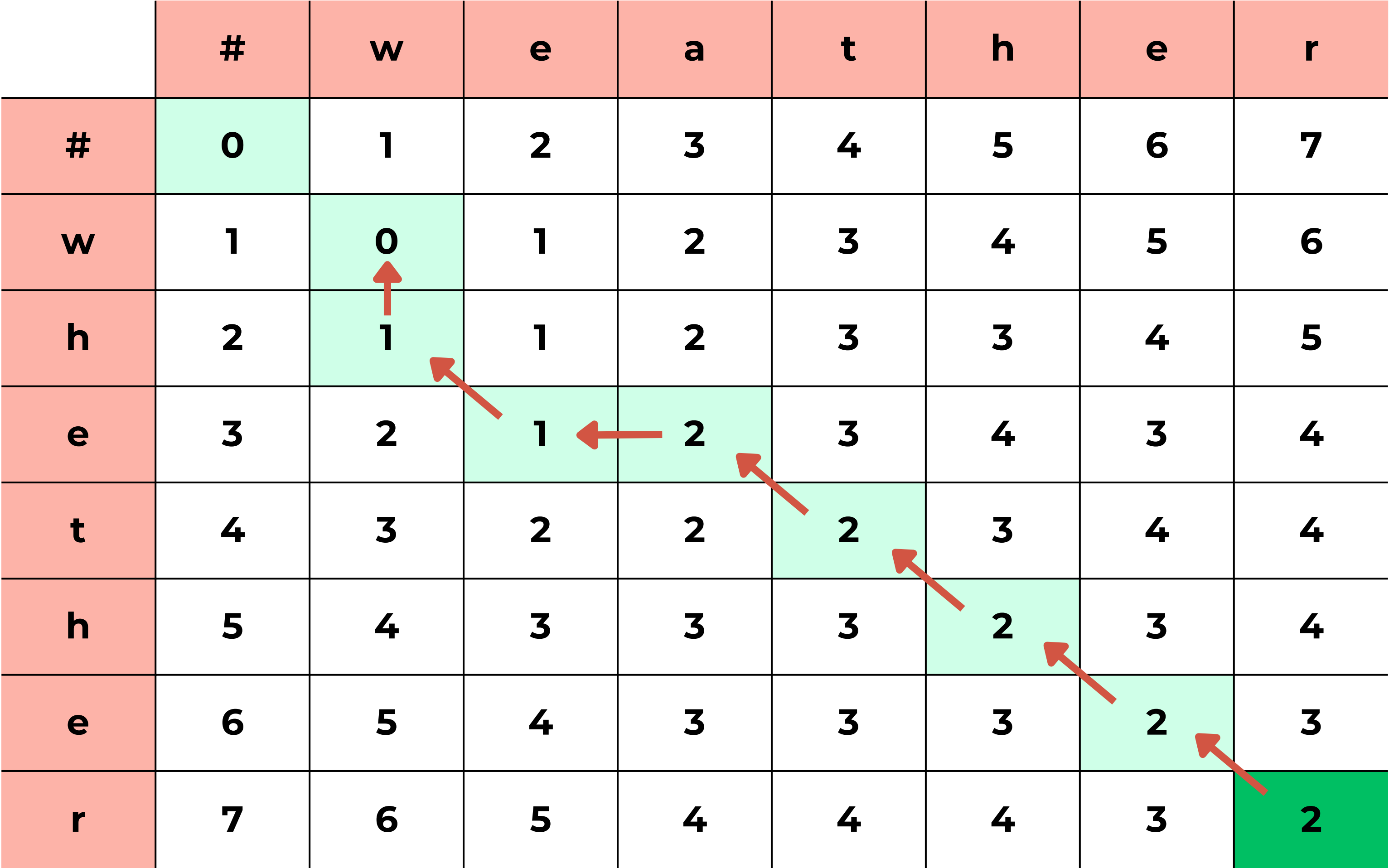
We found a Levenshtein distance of 2.

But, what that number 2
means?

BackTrace?



	#	w	e	a	t	h	e	r
#	0	1	2	3	4	5	6	7
w	1	0	1	2	3	4	5	6
h	2	1	1	2	3	3	4	5
e	3	2	1	2	3	4	3	4
t	4	3	2	2	2	3	4	4
h	5	4	3	3	3	2	3	4
e	6	5	4	3	3	3	2	3
r	7	6	5	4	4	4	3	2



weather $\longrightarrow$ **whether**

1. delete a[idx2]

we**a**ther $\longrightarrow$ wether

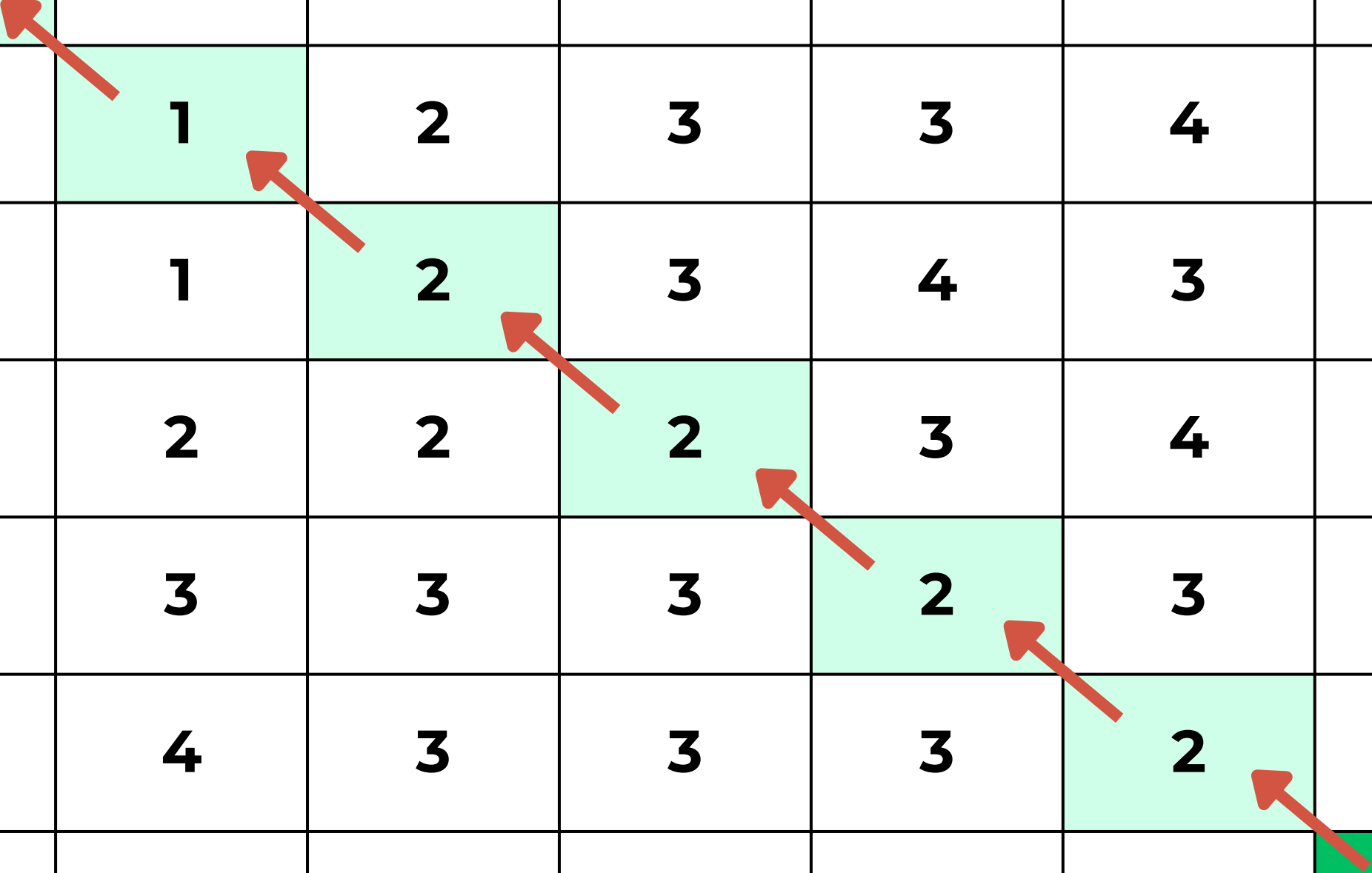
2. add h[idx1]

wether $\longrightarrow$ w**h**ether

Note: 2 operations.

OR

	#	w	e	a	t	h	e	r
#	0	1	2	3	4	5	6	7
w	1	0	1	2	3	4	5	6
h	2	1	1	2	3	3	4	5
e	3	2	1	2	3	4	3	4
t	4	3	2	2	2	3	4	4
h	5	4	3	3	3	2	3	4
e	6	5	4	3	3	3	2	3
r	7	6	5	4	4	4	3	2



weather $\longrightarrow$ **whether**

1. replace a[idx2] $\rightarrow$ e

we**a**ther $\longrightarrow$ we**e**ther

2. replace e[idx1] $\rightarrow$ h

w**e**ether $\longrightarrow$ wh**e**ther

Note: 2 operations.

The last two tables give the same distance, but they appear different? You saw that, although the distances are the same, the operations differ slightly.

This is not always the case.

References

https://youtu.be/We3YDTzNXEk?si=edCHphCsXi_mni1b

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End of out slide...

Thank you.